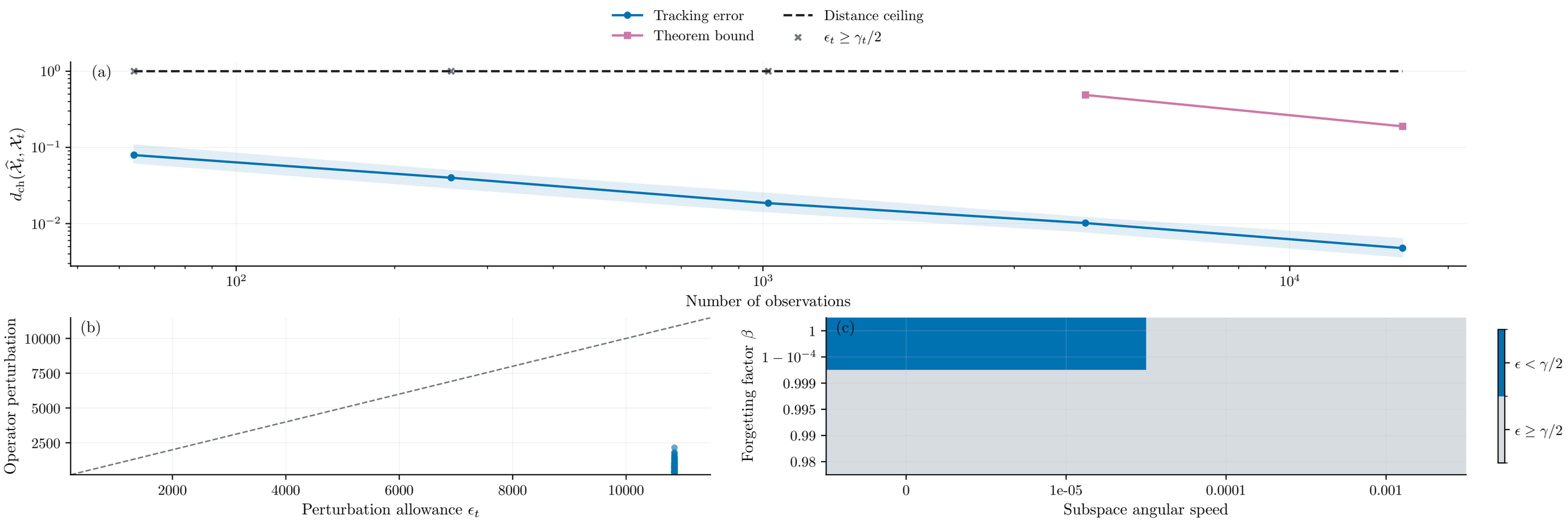

Tracking error
Theorem bound
Distance ceiling
$\epsilon_t \geq \gamma_t/2$
(a)
$d_{\mathrm{ch}}(\widehat{\mathcal{X}}_t, \mathcal{X}_t)$
10^0
10^{-1}
10^{-2}
10^2
10^3
10^4
Number of observations
(b)
Operator perturbation
10000
7500
5000
2500
2000
4000
6000
8000
10000
Perturbation allowance ϵ_t
(c)
Forgetting factor β
1
$1 - 10^{-4}$
0.999
0.995
0.99
0.98
0
1e-05
0.0001
0.001
Subspace angular speed
$\epsilon < \gamma/2$
$\epsilon \geq \gamma/2$